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Evaluation of a Culturally Tailored Mobile Text Messaging Intervention to Promote the Receipt of Pap-Test and HPV Vaccine

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Overview of the Research

Background

- Cervical cancer is one of the most preventable cancers, however, it presents a major health disparity among Asian immigrant women. Korean immigrant women, in particular, have a higher cervical cancer burden than other racial/ethnic groups (Gor, Chilton, Gamingue, & Hajek, 2011; Kagawa-Singer & Pourat, 2000). However, little research has been conducted to promote Pap test uptake and HPV vaccination among Korean immigrant women.

- A text message-based intervention was developed to improve cervical cancer screening and HPV vaccination, a *mScreening* project.

mScreening
: 7 days text messaging program
: Introducing cervical cancer; introducing the Pap test; introducing clinic, cost, and health professionals; introducing HPV vaccination; cost of HPV vaccination and review; assessing personal cultural-based barriers

Research Purpose

- This study is to evaluate a culturally tailored mobile text messaging intervention (*mScreening*): (1) content of *mScreening* and (2) method and format of the intervention delivery

Methods

- Data was collected through qualitative research methods including semi-structured interviews, three focus groups, and follow-up interviews with 30 young Korean immigrant women who attended and completed the *mScreening* intervention from April of 2011 to November of 2012.

Semi-structured interviews
- One week after the completion of the *mScreening* completion
- Less than 30mins
- Note taking

Focus groups
- A month after the *mScreeening* completion
- Between 1 hour and 1.5 hours
- Recorded
- $20

Follow-up interviews
- Three months later after the *mScreening* completion
- Less than 15mins
- Phone call
- $10

- Braun and Clarke' six phases of thematic analysis was used to identify core themes (Braun & Clarke, 2006).

Findings

1. Contents

1) Motivations to get a Pap-test or HPV vaccination: increased awareness and knowledge on prevention and early detection of cervical cancer, reduced fear and anxiety towards a Pap-test, and changes of thinking to cultural beliefs/attitudes

"I was reluctant to get Pap-test because I had a sort of negative feeling and prejudices toward women's clinic but my thoughts has been changed after I participated in this study. I started questioning the necessity of visiting women's clinic, so I planned to visit women's clinic in Korea while I was there during vacation. However, I could not make it. My mom disagreed because I am a single and young woman"

2) Increase awareness of preventive health orientation

"It was a good chance to check my health status as well as learn about cervical cancer related knowledge such as the causes of cervical cancer and how to prevent it. I rarely pay attention to my health because I am young and healthy. However, this program made me think about my health and preventive health orientation. I do not have a specific plan to have Pap-test or HPV vaccine but I would like to have one in the near feature."

3) Suggested contents for future intervention studies (e.g., other ways to prevent cervical cancer beyond HPV vaccine and Pap-test; HPV vaccine side effects; Information about medical bill; Interpreter services)

"I think that the messages have limited information. It might be helpful if you give us the summary of all messages with additional resources via emails, so we can get more information. I tried to search but I did not know which website I need to visit. In addition, I do not trust all the information that I found in the internet."

2. Method and Format

1) Text messaging along with visual images as effective to deliver cancer related health information

"Text messaging seems very useful and convenient. I do not need to search the internet or read books to learn about Pap-test and HPV vaccination. I was able to read the message on the bus or break time while I was on the program."

"It was a good chance to learn about cervical cancer and related knowledge. I did not feel bored while I was in the program because of messages with images and graphs. The messages were easy to understand."

2) Suggested solutions for technical issues and challenges: smart phone vs. regular phone, network connection issue, and message interval

"Even we marked our preferred time, our schedule could be changed. I am wondering that there are ways that each person choose starting time as well as interval each day? If you just send a link to access the program, we can do it when we have time with preferred interval. Then, we do not need to feel pressure or feel behind."





Conclusion & Implication

- **Contents:** Improved motivations to uptake of Pap-test or HPV vaccination and awareness of preventive health orientation
- Implications: (1) suggest multilevel intervention by including parents /community in the study, (2) include/expand the mScreening to promote other preventive diseases and immunization, (3) provide more information using various sources including blog, SNS, and reliable online resources

- **Methods and Format**: A text message-based intervention along with visual images could be effective tools to deliver tailored cancer health information.
- Implications: (1) suggest to use smartphone to develop app to overcome the limitation of message length and improve visual message delivery, (2) add the function of choosing next message by the participant to improve participant engagement (to resolve message interval issues)

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